

Equivalence Completion and Forced Collapse of Charged–Current Degrees of Freedom

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Abstract

We identify a forced tensor-level relation implied by admissibility, standing conservation, and closure. Apparent independent parameters governing charged–current interactions in renormalizable electroweak bookkeeping are shown to overcount tensor load. Explicit cross-skin transport to an admissible low-energy description yields a unique invariant witness, forcing elimination of the apparent independence. The result is eliminative, invariant under admissible redescription, and introduces no new physics.

1 Apparent Degrees of Freedom in Renormalizable Bookkeeping

In renormalizable electroweak bookkeeping, charged–current interactions are mediated by a massive vector boson with coupling g_2 and symmetry-breaking scale v , with

$$m_W = \frac{1}{2}g_2v. \quad (1)$$

Within this skin, g_2 and v are treated as independent parameters. Charged–current amplitudes are computed by explicit W exchange.

2 Admissible Low-Energy Skin

An admissible low-energy description of the same charged–current physics is given by a four-fermion effective interaction

$$\mathcal{L}_{\text{CC}}^{\text{EFT}} = -\frac{G_F}{\sqrt{2}}(\bar{\psi}_1\gamma^\mu(1-\gamma^5)\psi_2)(\bar{\psi}_3\gamma_\mu(1-\gamma^5)\psi_4), \quad (2)$$

characterized by a single coefficient G_F . This skin contains no explicit gauge coupling or mediator mass.

3 Lemma 1: Explicit Cross-Skin Transport

Integrating out the W boson at tree level yields

$$\mathcal{L}_{\text{eff}} = -\frac{g_2^2}{8m_W^2}(\bar{\psi}_1\gamma^\mu(1-\gamma^5)\psi_2)(\bar{\psi}_3\gamma_\mu(1-\gamma^5)\psi_4). \quad (3)$$

Substituting $m_W = \frac{1}{2}g_2v$ gives

$$\frac{g_2^2}{8m_W^2} = \frac{1}{2v^2} = \frac{G_F}{\sqrt{2}}. \quad (4)$$

Thus admissible transport identifies the charged–current tensor load with the single invariant

$$G_F = \frac{1}{\sqrt{2}v^2}. \quad (5)$$

4 Lemma 2: Invariant Witness Counting

Let a tensor load admit n independent degrees of freedom. Under admissible redescription, each independent degree must correspond to an invariant witness in every admissible skin describing the same regime. If an admissible skin exposes fewer invariant witnesses, then the higher-dimensional parameterization contains redundancy rather than additional tensor structure.

5 Failure of Independence

Assume g_2 and v constitute two independent tensor degrees of freedom for charged–current interactions. By Lemma 2, two invariant witnesses must survive admissible transport. The low-energy skin exposes only one invariant witness, G_F . Preservation of independence would require either loss of tensor load, hidden un-witnessed structure, or representational privilege. Each constitutes standing failure.

6 Forced Tensor Relation

The assumption of independence is inadmissible. The charged–current tensor load is one-dimensional and is fully specified by G_F . The parameters g_2 and v cannot both be independently physical for this sector. Any admissible skin exposing both necessarily contains redundancy.

7 Corollaries

- Charged–current interaction strength admits a unique invariant specification.
- Mediator-based and contact-interaction descriptions are tensor-identical after redundancy elimination.
- No admissible extension permits independent variation of charged–current strength at fixed G_F without introducing forbidden tensor load.

Conclusion

Equivalence completion under admissibility and closure forces collapse of apparent freedom in the charged–current sector. The result is eliminative, representation-invariant, and irreversible.